Kimberly Palaguachi-Lopez

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EDUCATION

Columbia University Mailman School of Public Health – *New York, NY* **May 2026**
Master of Science in Biostatistics | Health Data Science Track

- Courses: Biostatistics I & II, Probability, Statistical Inference, SAS programming, Data Science I & II, Survival Analysis, Longitudinal Data Analysis, Randomized Clinical Trial Methodology, SQL: Relational Databases, Consulting Seminar.

Lehigh University – *Bethlehem, PA* **May 2024**
Bachelor of Science in Population Health | Biostatistics Concentration

- Relevant Coursework: Population Health Research Methods, Methods in Epidemiology, Population Health Data Science & Algorithms Labs I & II, Spatial Epidemiology, Environmental Epidemiology, Outbreak Science & Disease Forecasting, Application of Statistical Inference.

STATISTICAL & DATA ANALYSIS RESEARCH EXPERIENCE

Student Analyst January 2026 – Present
Naomi Berrie Diabetes Center | Columbia University Irving Medical Center (CUMC) New York, NY

- Performed multivariate analysis of proteomic and lipid biomarker data integrated with longitudinal clinical trial data from the LIFE MOMs gestational diabetes cohort, including principal component analysis (PCA), unsupervised clustering, and biomarker-driven data visualization.
- Prepare statistical summaries, tables, figures, and outputs supporting clinical insights and translational research.
- Partnered with clinical investigators to interpret results and support statistical manuscript sections development.

Clinical Data Science – Statistics Co-op August 2025 – February 2026
Boehringer Ingelheim Pharmaceuticals Inc. | Biostatistics and Data Science Department Remote

- Supported clinical validation and statistical analyses across phase II studies by analyzing clinical databases using SDTM and ADaM datasets to generate accurate, quality-controlled tables, listings, and figures (TLFs) from SAS to R.
- Executed data cleaning and standardization into a regulatory compliant ADaM standard dataset for documentation.
- Programmed and validated descriptive statistics and endpoint summary tables in R, documenting assumptions, derivations, and QC checks within an Analysis Data Reviewers' Guide and mock ADS plan.
- Developed interactive user-friendly R Shiny interactive dashboards for clinical quality monitoring and visualize key data outcomes for cross-functional teams and CRO partners, improving efficiency for internal study teams.

Research Intern – Biostatistics & Clinical Research June 2025 – August 2025
Joslin Diabetes Research Center | Harvard Medical School Boston, MA

- Analyzed NIH phase II trial using SDTM datasets with an intention-to-treat (ITT) analysis approaches and linear mixed-effect models in SAS to evaluate longitudinal kidney function outcomes in Type 1 Diabetes patients for PI.
- Synthesized statistical results into tables, figures, and manuscript-ready summaries, supporting clinical insights.
- Presented key clinical insights on treatment effect in kidney function; generating evidence to inform future large-scale research studies and population health interventions aimed at delaying kidney decline in T1D.

Improving Lives Research Internship June 2024 – August 2024
Breakthrough T1D Research Foundation | Formerly JDRF New York, NY

- Conducted a clinical trial literature review and landscape analysis of 200+ studies in Type 1 and Type 2 Diabetes cardiovascular disease, identifying gaps in biomarker utilization and endpoint selection.
- Supported study design consideration by identifying gaps in endpoint selection and biomarker utilization relevant to cardiovascular and heart failure outcomes through surrogate endpoints.
- Synthesized findings into a formal research presentation on the landscape analysis to the research department and proposing three promising cardiovascular drug treatments for future grant application strategies.

Epidemiology Statistical Analysis Assistant
College of Health, Lehigh University Januaray 2024 – May 2024

- Processed and merged 300+ participant baseline and follow-up data, and performed exploratory analysis.

- Conducted descriptive statistics and performed hypothesis testing, identifying significant changes in health scales over time, by analyzing cross-sectional survey responses.
- Co-authored a manuscript selected for presentation at APHA 2024 conference contributing to public health discourse, by synthesizing findings and collaborating with faculty on statistical interpretation.

Research Student June 2023 – September 2023
STEM – Summer Institute at Lehigh University Bethlehem, PA

- Processed 12 years (2011-2023) of Mortality Morbidity Weekly Report (MMWR) Influenza A/B data across 67 Pennsylvania counties identifying seasonal flu trends, by developing heat maps for data visualization using Python.
- Built an automated data pipeline to streamline CDC data to epidemiological time-series modeling workflow and estimated parameter values for an Agent-Based Model (ABM) simulated in Netlogo.
- Evaluated ABM predictive accuracy using model validation, sensitivity analysis, and scenario simulation to inform vaccine rollout strategies in high-risk populations.

Co-Research Author September 2022 – December 2022
College of Health at Lehigh University Bethlehem, PA

- Conducted extensive literature review research on the relationship between asthma and cold temperatures.
- Contributed to the drafting and publication of an academic research paper focused on nasal malondialdehyde mediating the effects of cold air exposure on pediatric asthma symptoms.

Undergraduate Data Analyst Assistant February 2021 – May 2021
Institute of Critical Race & Ethnic Studies at Lehigh University Bethlehem, PA

- Attended weekly 3 hours of meetings with colleagues to review agendas and planning new projects.
- Gather 5,000+ participant's personal experiences into Google Sheets during the COVID-19 pandemic via social media, used to determine how the pandemic is affecting race and gender in data analysis.

WORK EXPERIENCE

Village Care Max – Manhattan, NY June 2023 – August 2023
Data Entry Intern

- Retrieved vaccination Electronic Health Record (EHR) data from the Citywide Immunization Registry for Village Care patients across the broader New York City area.
- Systematically processed and incorporated new data into database systems, cross-referenced and validated against Village Care records.
- Proficiently managed, organized, and securely stored over 3,000 data entries within spreadsheets, consistently upholding strict HIPAA compliance standards.

Pathways To Trust – Remote July 2022 – August 2022
Strategic Partnership Donor Development Intern

- Conducted comprehensive research to identify prospective donors and sponsors for Pathways To Trust, facilitating the organization's patient advocacy programs.
- Collaborated with PTT partners to actively seek out suitable research centers for disease clinical trials, contributing to the expansion of the organization's participant network.
- Orchestrated effective communication and teamwork with a group of interns to proactively explore grant opportunities for PPT to ensure eligibility for future funding.

PUBLICATIONS

Health Effects of a Natural Experiment Introducing Nature Window Views Among Healthcare Workers
Submitted to Health Environments Research & Design Journal November 2025
16 Pages, Co- authors: Fry, Dustin; Ganns, Elyse; Palaguachi-Lopez, Kimberly; Eaves-Lyter, Emily; McIntire, Russell

Nasal Malondialdehyde Mediating the Effects of Colder Air Exposure on Pediatric Asthma Symptoms
Pediatric Research April 2024
23 Pages, Co- authors: He, Linchen; Norris, Christina; Palaguachi-Lopez, Kimberly; Barkjohn, Karoline; Li, Zhen; Li, Feng; Zheng, Yinping; Black, Marilyn; Bergin, Michael; Zhang, Junfeng (Jim)

CONFERENCE ABSTRACTS

Are Greener Window views associated with better mental health among workers at a rehabilitation hospital? A natural experiment

APHA 2024 Annual Meeting and Expo November 2024

Co-authors: McIntire, Russell; Palaguachi-Lopez, Kimberly; Rafai, Nader; Lyter, Emily; Kolakowsky-Hayner, Stephanie

LEADERSHIP ACTIVITIES

R.I.S.E Mentorship Columbia University – New York, NY August 2025 – Present

Mentor

- Mentoring three first-year M.S. Biostatistics students to support their transition into graduate-level coursework, statistical programming, and research opportunities.
- Facilitating connections to faculty-led research projects, departmental seminars, and professional development workshops to enhance mentees' academic growth and research engagement.
- Collaborating with program administrators to monitor mentees' progress and tailor mentorship strategies to academic and professional goals.

HONORS & AWARDS

Dean's List – *Lehigh University* Spring 2021- Spring 2024

Recognized outstanding academic achievement who have earned a scholastic average of 3.6, or greater; and carry at least 12 hours of regularly graded courses.

Hispanic Scholarship Fund Scholar June 2024

Selected as one of 10,000 scholars from a competitive pool of over 100,000 applicants. The distinction recognizes academic excellence, hard work, and dedication.

Tri Alpha Honor Society Member – *Lehigh University* March 2024

The Alpha Alpha Alpha (Tri-Alpha) Epsilon Chapter at Lehigh recognizes the academic achievements of first-generation college students who have achieved an overall undergraduate GPA of at least 3.5 on a 4.0 scale.

SKILLS

Statistical Software: Proficient in R, SAS, Python, and SQL for relational databases, Git/GitHub & Posit for collaboration.
Clinical Research Methods: Design and analysis of phase I–III, endpoint selection, and statistical analysis plans (SAPs); interim analyses, group sequential methods, power and sample size calculations, and mixed-effects modeling.
Statistical Analysis: Survival analysis, regression modeling (linear, logistic, Cox), longitudinal & multivariate analysis.
Machine Learning: Supervised & unsupervised learning, classification, regression, clustering, dimensionality reduction, ensemble methods, support vector machines, regularization, cross-validation, feature engineering.
Data Management: Data cleaning, integration, and quality control for clinical research dataset; CDISC SDTM/ADaM.
Data visualization: SAS GRAPH, R packages (ggplot2, plotly, Tplyr, RShiny for interactive dashboards), Excel, ArcGISPro.